Data Preprocessing

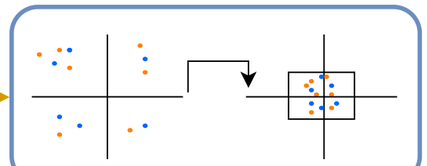
Dataset

Torque	Temp.	Tool
150		20
	70	16
200	80	

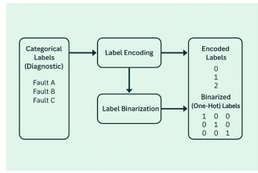
Imputation

Dataset

Torque	Temp.	Tool
150	75	20
175	70	16
200	80	18



Scaling



Label encoding

Available data

Training set | Test set

Train test split

MLP

Input → Dense(64, ReLU) → Dropout (0.3) → Dense(32, ReLU) → Dropout (0.2) → Dense(1, Sigmoid)

LSTM

Input → LSTM(32) → Dropout (0.3) → Dense(1, Sigmoid)

CNN-BiGRU

Input → Conv1D(64, ReLU) → MaxPool1D (pool=2) → Bidirectional (GRU(32)) → Dropout (0.3) → Dense(1, Sigmoid)

soft-voting (probability averaging)

Final Prediction